

Discrete Response, Time Series and Panel Data - Exercises (Rep.)

Nils Rechberger

2026-03-05

Series 1

Exercise 1.1

a

12

b

None

c

7

d

365

Exercise 1.2

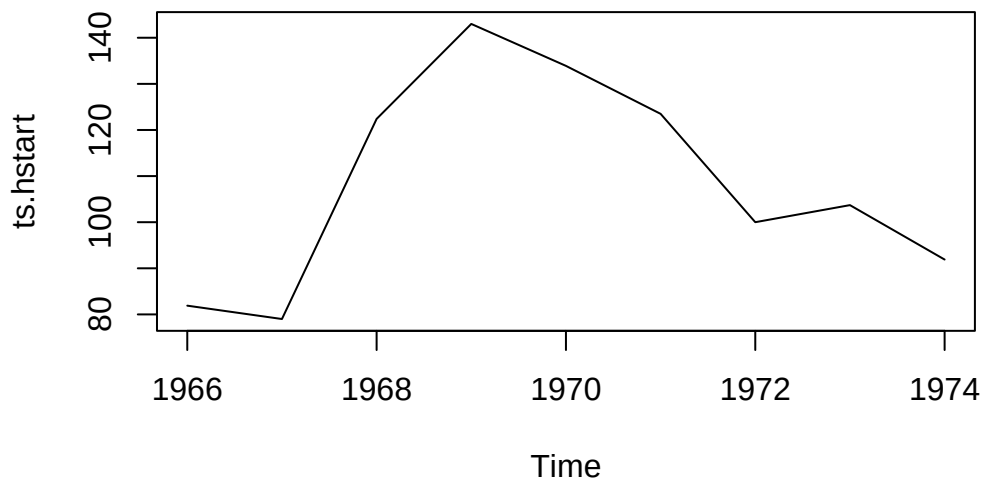
```

hstart <- read.table("./data/hstart.dat")

ts.hstart <- ts(
  data=hstart,
  start=1966,
  end=1974
)

plot(ts.hstart)

```



The process is not stationary. We see a seasonal trend from 1967 to 1972.

Exxercise 1.3

a

```

set.seed(1)

Et <- ts(rnorm(101, 0, 1))
Et[1] <- 0
y1 <- 0 #delete later

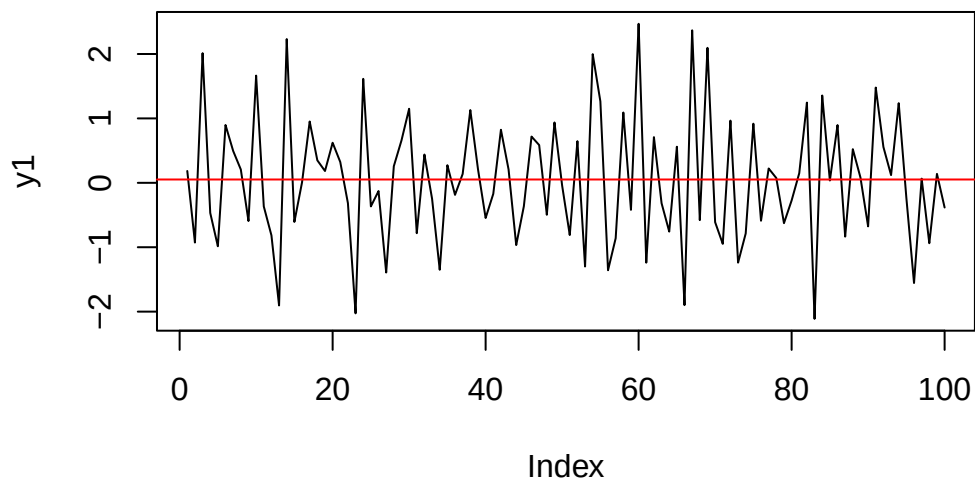
```

```

for (i in 2:length(Et)) {
y1[i] <- Et[i] - 0.5 * Et[i-1]
}

y1 = y1[2:length(y1)]
ts.y1 = ts(y1)
plot(y1, type="l")
abline(h=mean(y1), col="red")

```



The process seems stationary.

b

```

set.seed(1)

Et <- ts(rnorm(101, 0, 1))
Et[1] <- 0
y2 <- 0 #delete later

for (i in 2:length(Et)) {

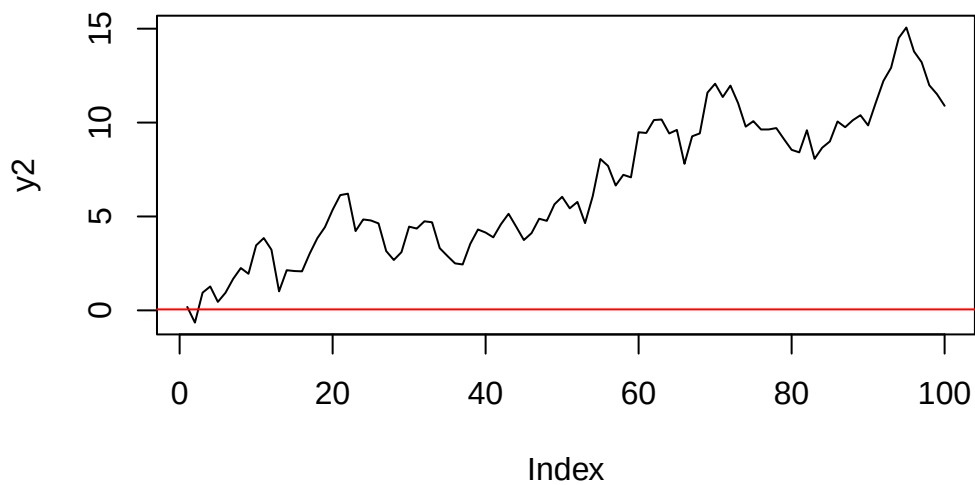
```

```

y2[i] <- y2[i-1] + Et[i]
}

y2 = y2[2:length(y2)]
ts.y2 = ts(y2)
plot(y2, type="l")
abline(h=mean(y1), col="red")

```



The process is not stationary.

c

```

set.seed(1)

Et <- ts(rnorm(101, 0, 1))
Et[1] <- 0
y3 <- 0 #delete later

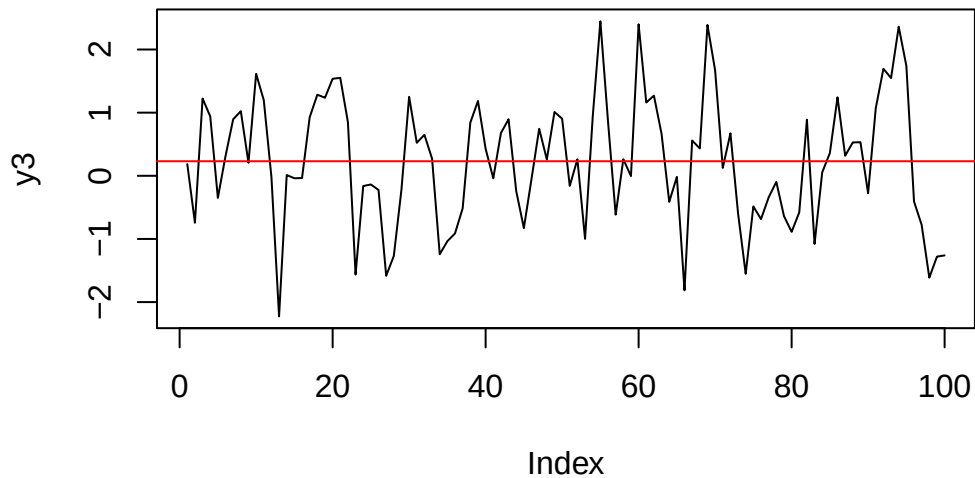
for (i in 2:length(Et)) {
  y3[i] <- 0.5 * y3[i-1] + Et[i]
}

```

```

y3 = y3[2:length(y3)]
ts.y3 = ts(y3)
plot(y3, type="l")
abline(h=mean(y3), col="red")

```



The process seems stationary.

d

```

set.seed(1)

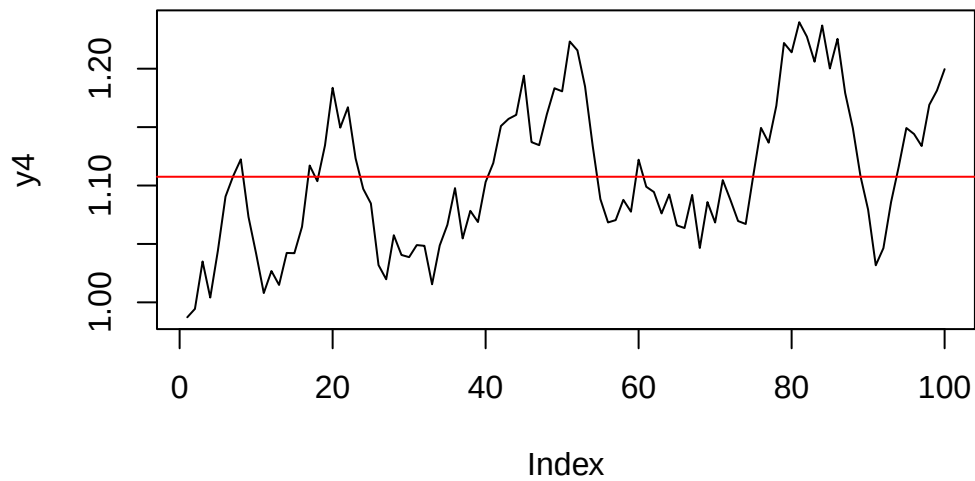
Et <- ts(runif(101, 0.95, 1.05))
Et[1] <- 0
y4 <- 1 #delete later

for (i in 2:length(Et)) {
  y4[i] <- y4[i-1] * Et[i]
}

y4 = y4[2:length(y4)]

```

```
ts.y4 = ts(y4)
plot(y4, type="l")
abline(h=mean(y4), col="red")
```



The model is not stationary

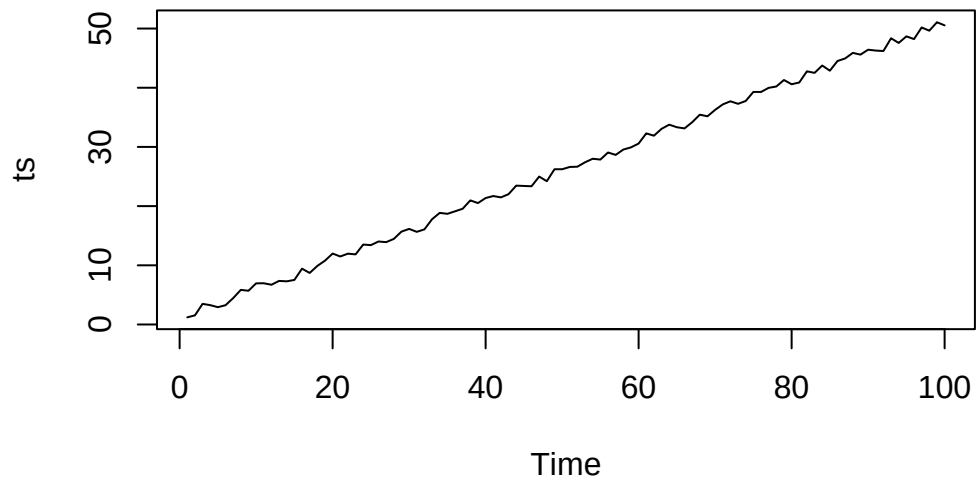
Series 2

Exercise 2.1

a

```
t <- seq(1, 100, length = 100)
data <- 0.5 * t + 1 + runif(100, -1, 1)
ts <- ts(data)

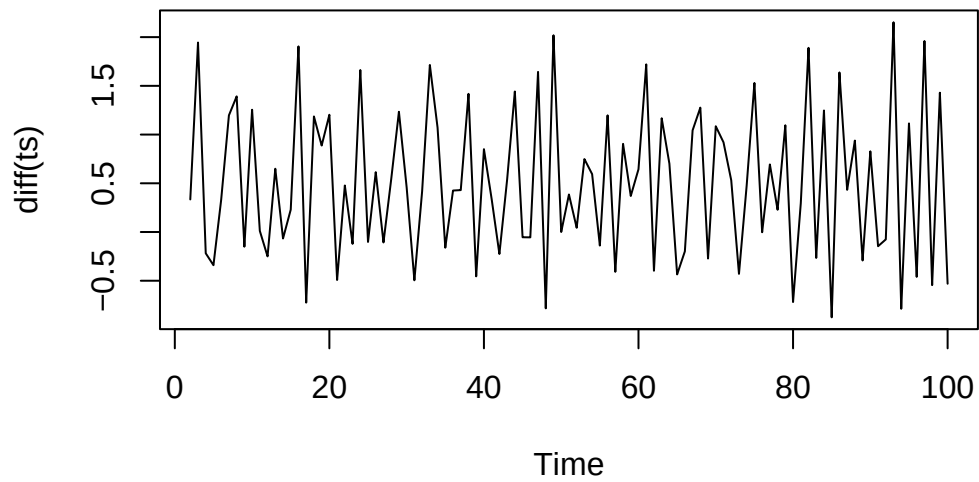
plot(ts)
```



```
print(length(ts))
```

```
[1] 100
```

```
plot(diff(ts))
```



```
print(length(diff(ts)))
```

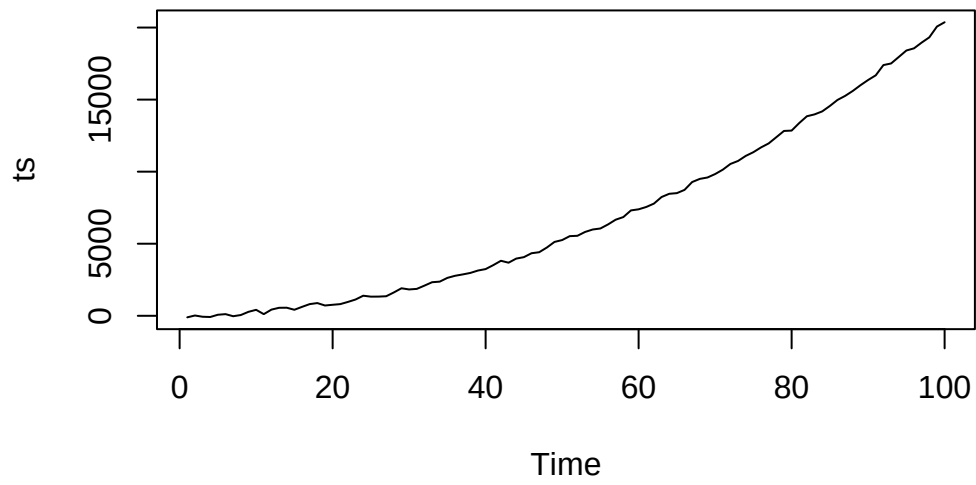
[1] 99

We can remove the linear trend with one time backward differencing. The length of the ts increases by 1.

b

```
t <- seq(1, 100, length = 100)
data <- 2 * t**2 + 3 * t - 1 + runif(100, -200, 200)
ts <- ts(data)

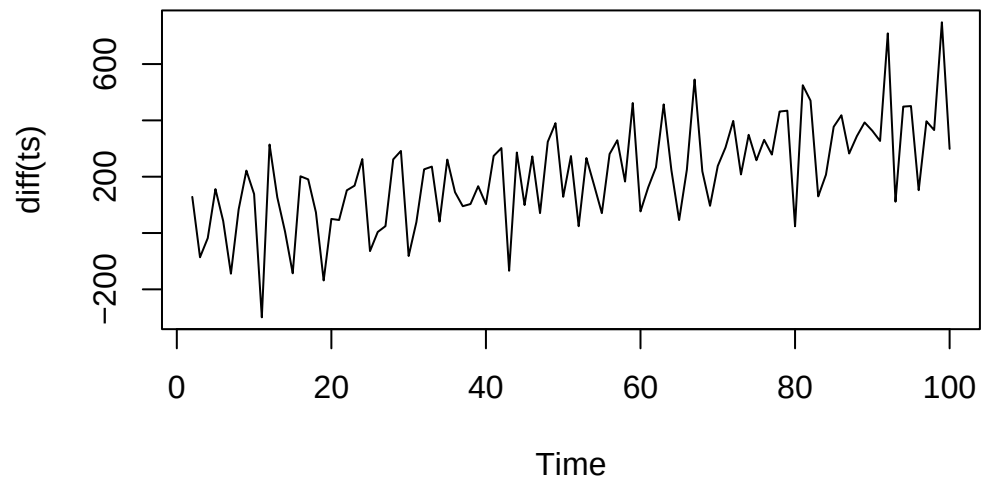
plot(ts)
```



```
print(length(ts))
```

```
[1] 100
```

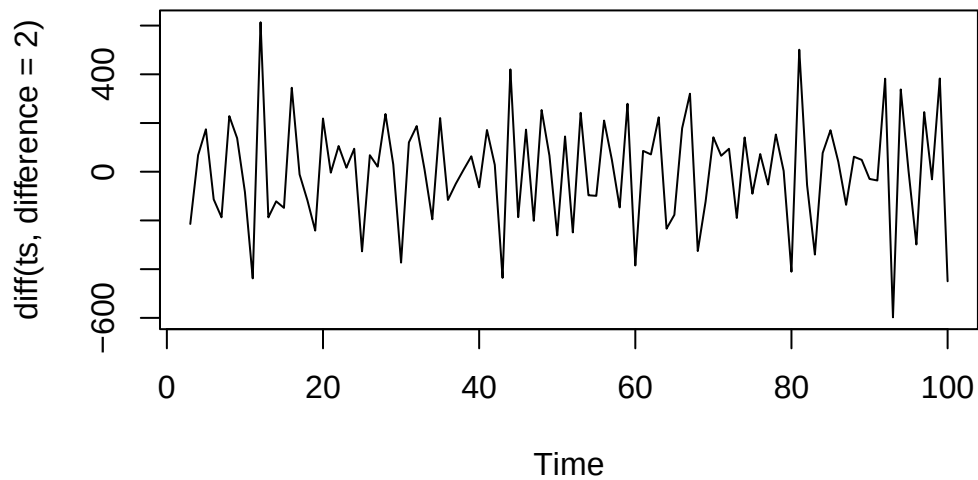
```
plot(diff(ts))
```



```
print(length(diff(ts)))
```

```
[1] 99
```

```
plot(diff(ts, difference=2))
```



```
print(length(diff(ts, difference=2)))
```

```
[1] 98
```

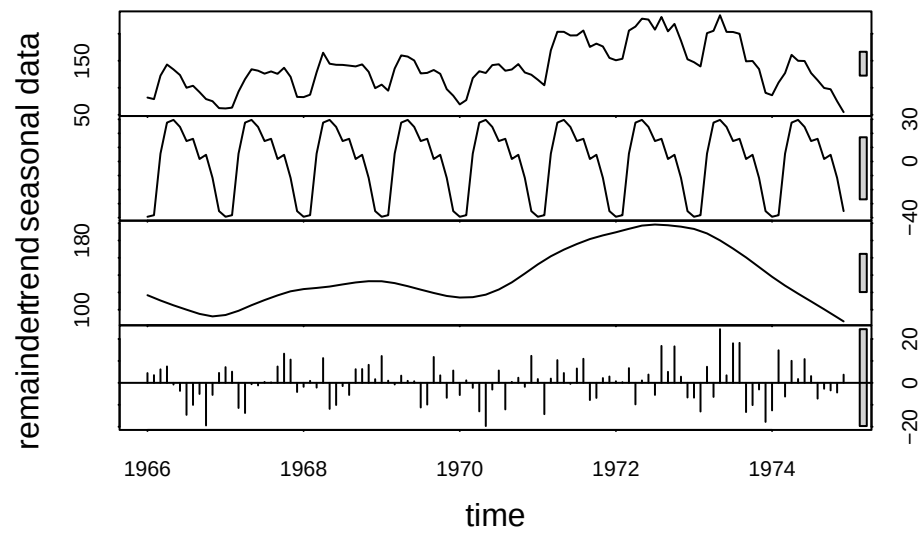
We can remove the linear trend with two times backward differencing. The length of the ts increases by 2.

Exercise 2.2

```
hstart <- read.table("./data/hstart.dat")

ts.hstart <- ts(
  data = hstart$V1,
  start = c(1966, 1), # January 1966
  end = c(1974, 12), # December 1974
  frequency = 12      # Monthly data
)

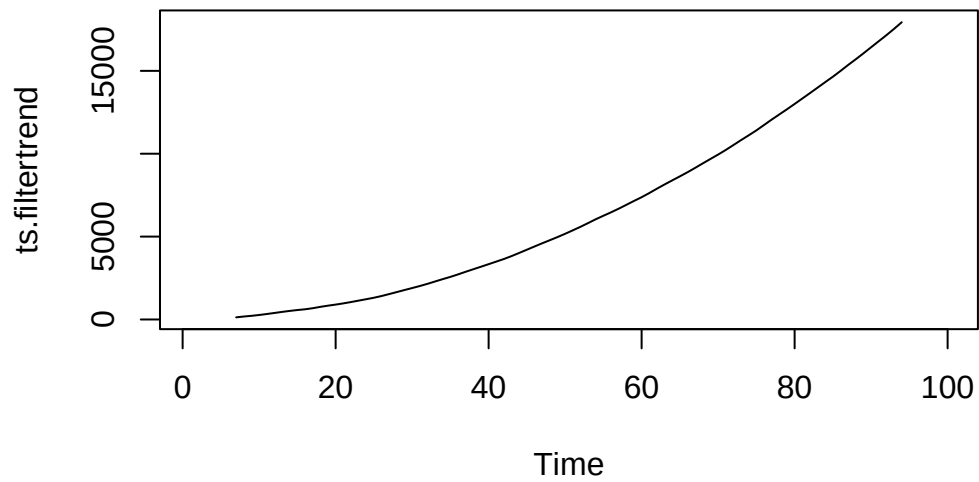
decomp <- stl(ts.hstart, s.window = "periodic")
plot(decomp)
```



b

```
weights <- c(0.5, rep(1, 11), 0.5) / 12
ts.filtertrend <- filter(
  ts,
  filter = weights,
  sides = 2
)

plot(ts.filtertrend)
```

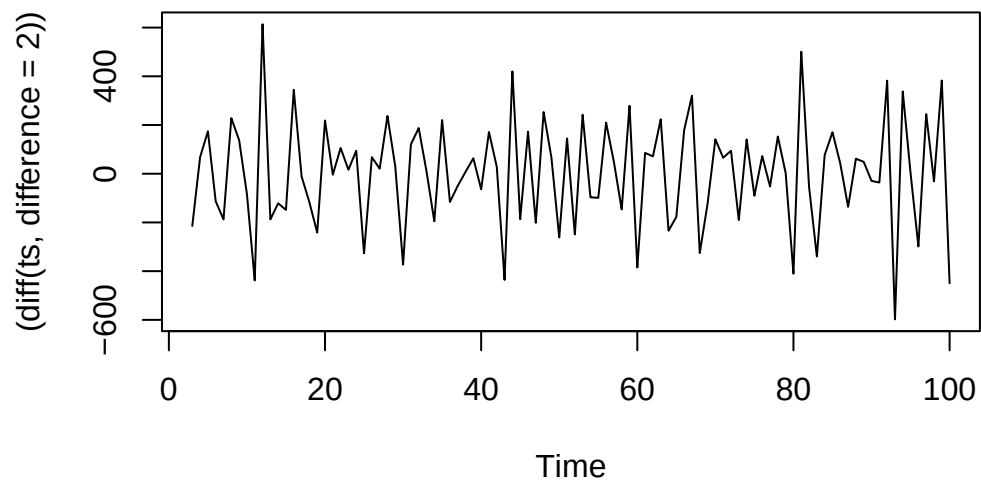


c

```
hstart <- read.table("./data/hstart.dat")

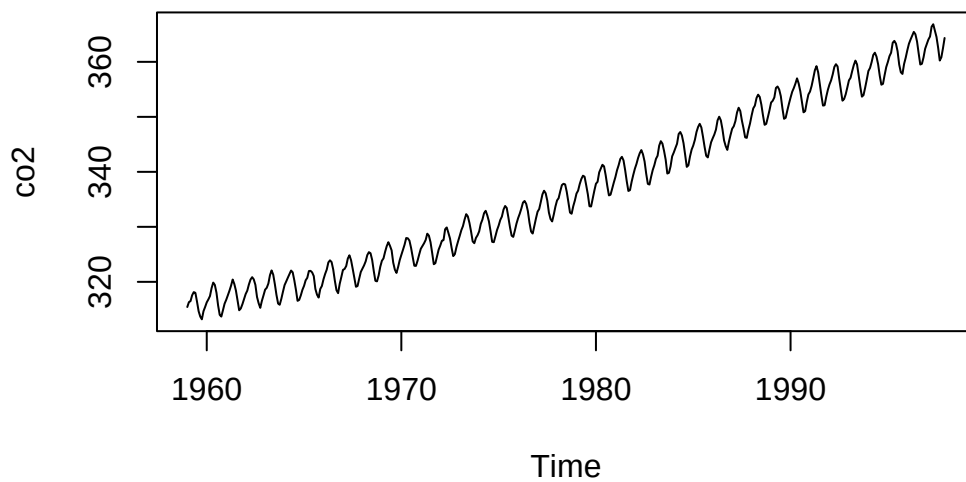
ts.hstart <- ts(
  data = hstart$V1,
  start = c(1966, 1), # January 1966
  end = c(1974, 12), # December 1974
  frequency = 12      # Monthly data
)

plot((diff(ts, difference=2)))
```

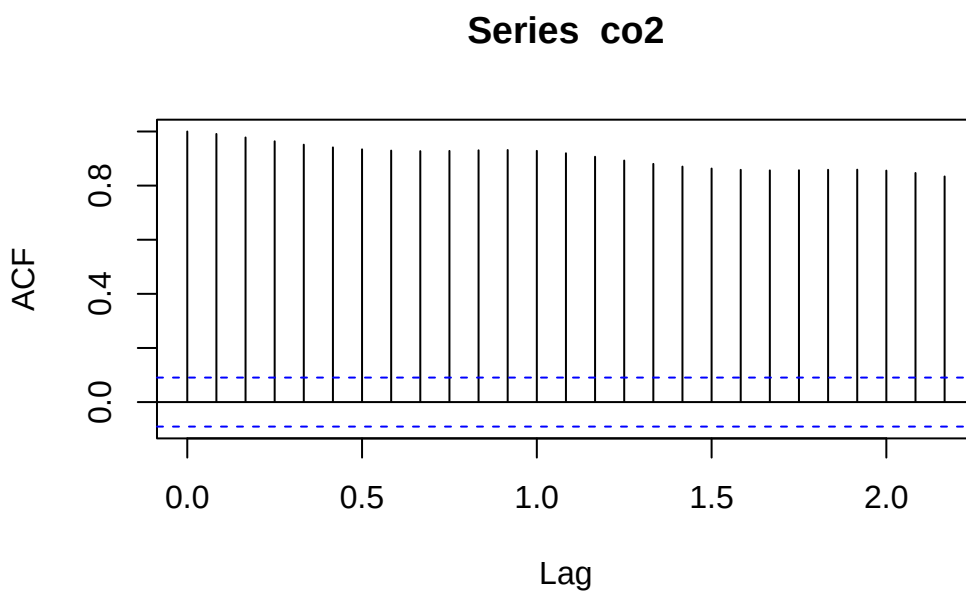


Exercise 2.3

```
data(co2)
plot(co2)
```



```
acf(co2)
```



Series 3

Exercise 3.1

A -> 1 B -> 3 C -> 2 D -> 4

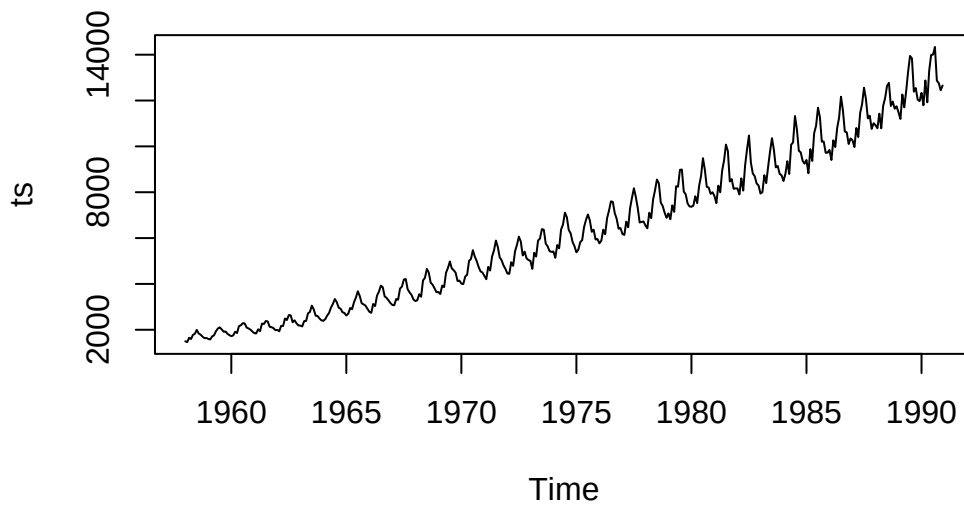
Exercise 3.2

a

```
cbe <- read.table("./data/cbe.dat", header=TRUE)

ts <- ts(
  data=cbe$elec,
  start=c(1958, 1),
  end=c(1990, 12),
  frequency=12
)

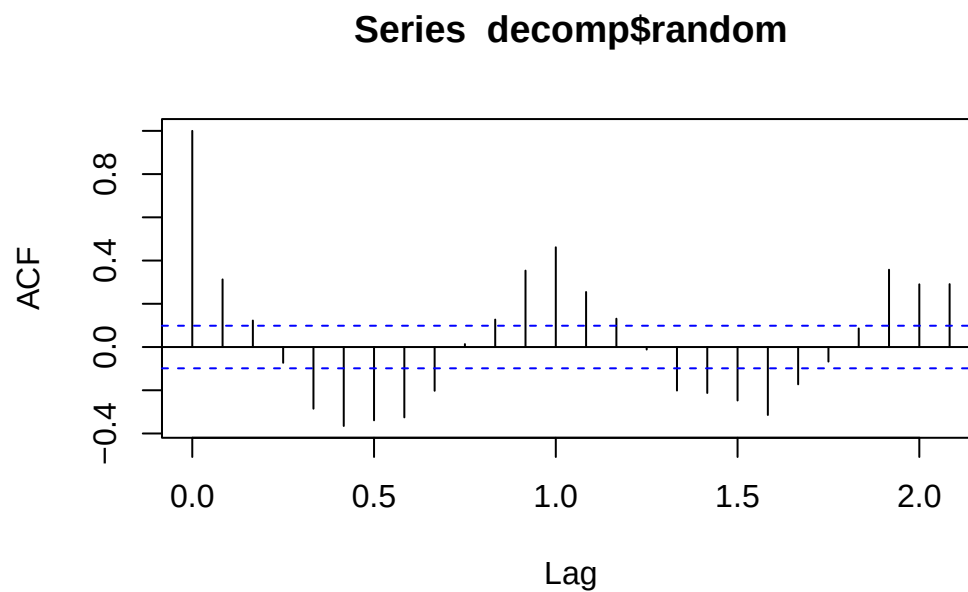
plot(ts)
```



It is not meaningful to interpret the correlogram of this time series because the process is not stationary.

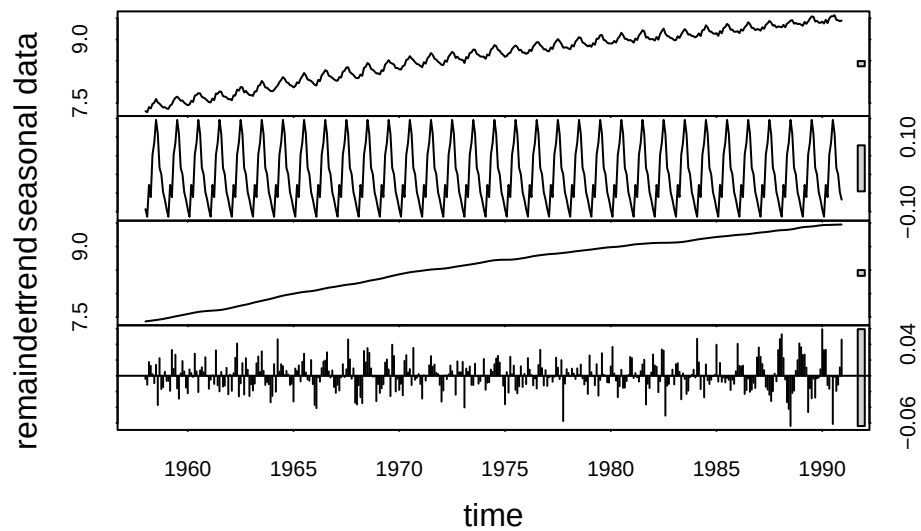
b

```
decomp <- decompose(ts, type = "multiplicative")  
acf(decomp$random, na.action = na.pass)
```

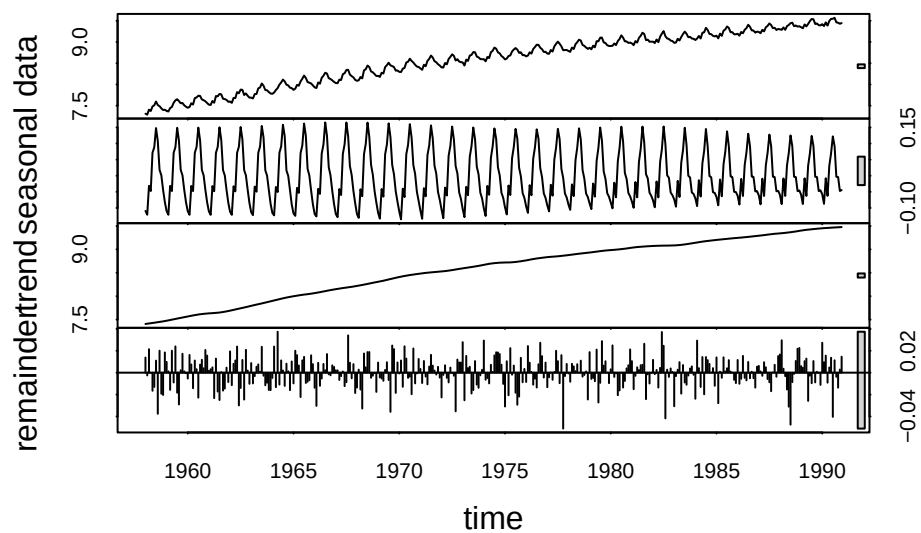


c

```
elec.stl1 <- stl(log(ts), s.window="periodic")  
plot(elec.stl1)
```



```
elec.stl2 <- stl(log(ts), s.window=13)
plot(elec.stl2)
```



By simply log-transforming the data does make the process stationary.

Series 4

Exercise 4.1

a

```
data <- arima.sim(list(ar = c(0.8)), n = 1000)
```

b

```
ARMAacf(ar=c(0.8), lag.max = 10)
```

0	1	2	3	4	5	6	7
1.0000000	0.8000000	0.6400000	0.5120000	0.4096000	0.3276800	0.2621440	0.2097152
8	9	10					
0.1677722	0.1342177	0.1073742					

c

$$p(k) = a_1^k = 0.8^k$$

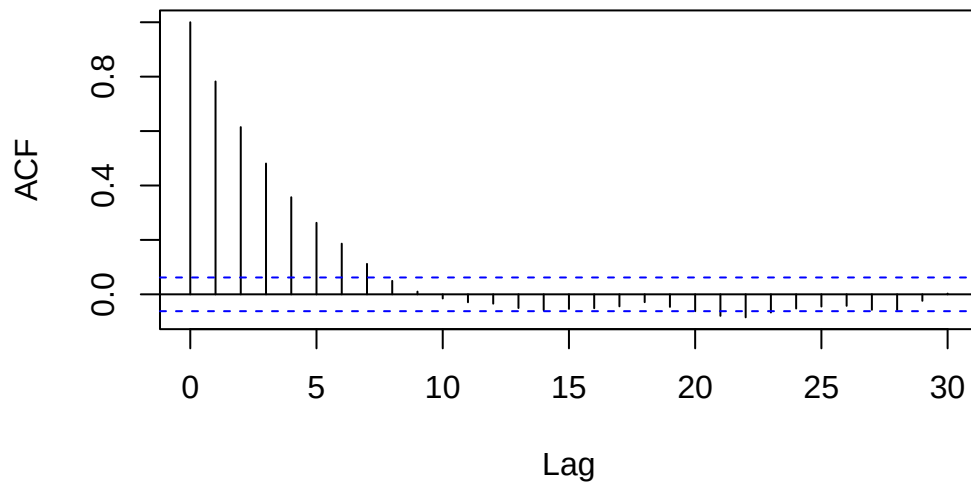
d

```
ARMAacf(ar=c(0.8), lag.max = 10)
```

0	1	2	3	4	5	6	7
1.0000000	0.8000000	0.6400000	0.5120000	0.4096000	0.3276800	0.2621440	0.2097152
8	9	10					
0.1677722	0.1342177	0.1073742					

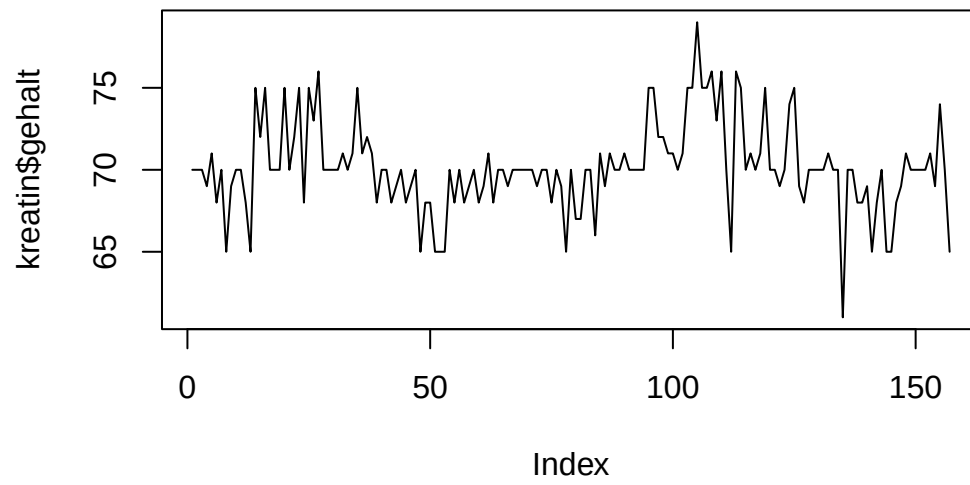
```
acf(data)
```

Series data



Exercise 4.2

```
kreatin <- read.table("./data/kreatin.dat", header=TRUE)
plot(kreatin$gehalt, type="l")
```

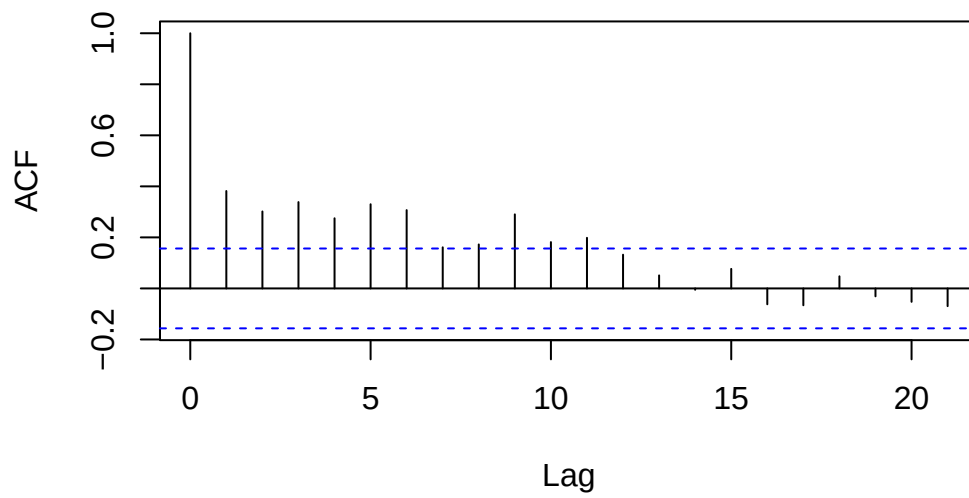


The process should follow white noise.

b

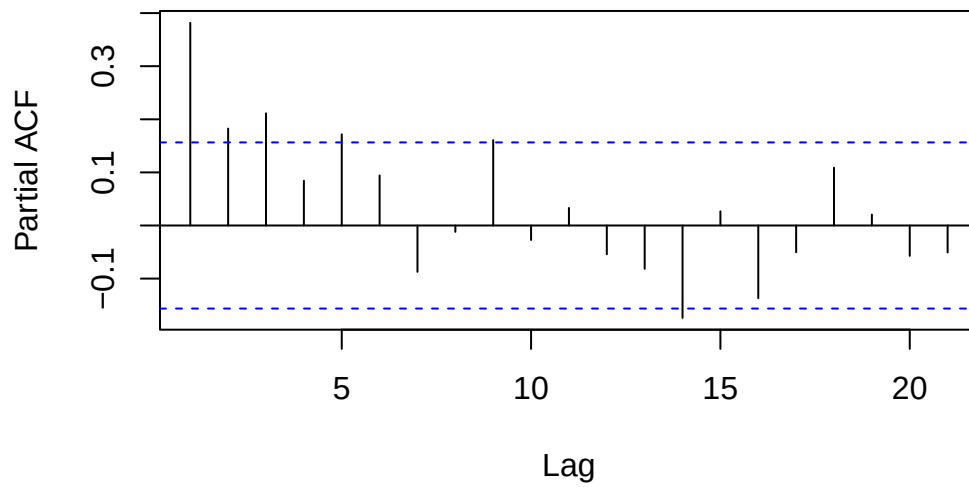
```
acf(kreatin$gehalt)
```

Series kreatin\$gehalt



```
pacf(kreatin$gehalt)
```

Series kreatin\$gehalt



White noise fits.

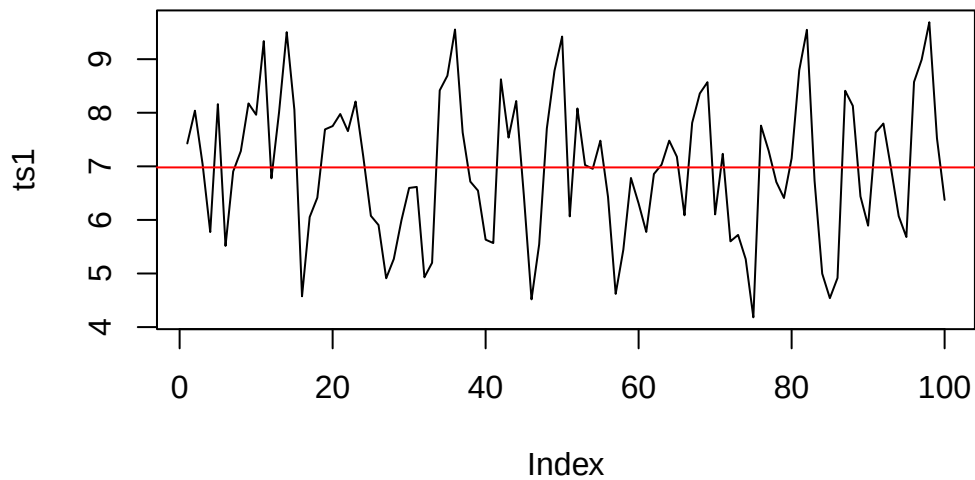
Exercise 4.3

```
data <- read.table("./data/ts_S3_A2.dat", header=TRUE)

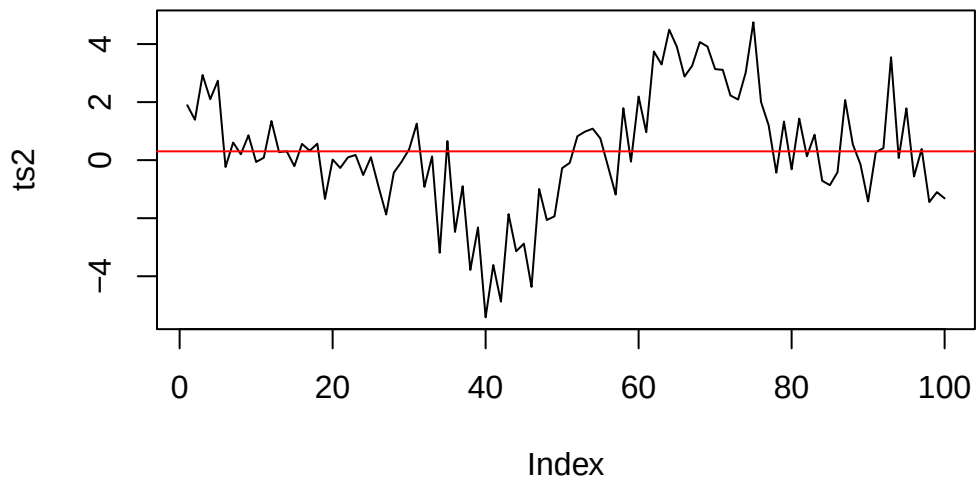
ts1 <- data$ts1
ts2 <- data$ts2
```

a

```
plot(ts1, type="l")
abline(h=mean(ts1), col="red")
```



```
plot(ts2, type="l")
abline(h=mean(ts2), col="red")
```



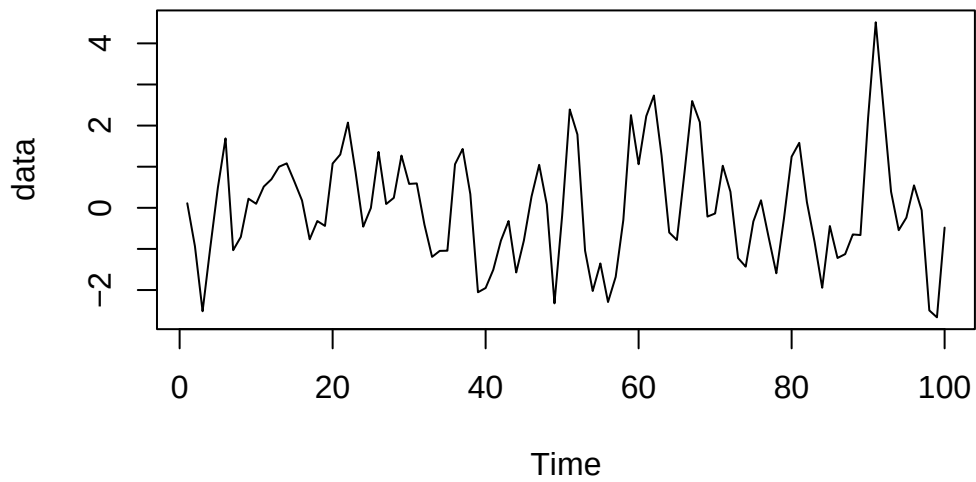
ts1 seems stationary, *ts2* not.

Series 5

Exercise 5.1

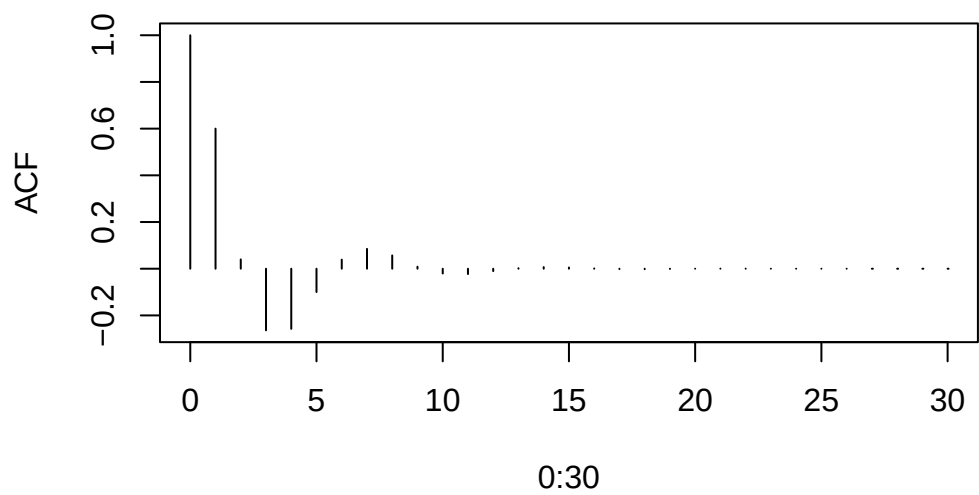
a

```
data <- arima.sim(list(ar = c(0.9, -0.5)), n = 100)
plot(data)
```

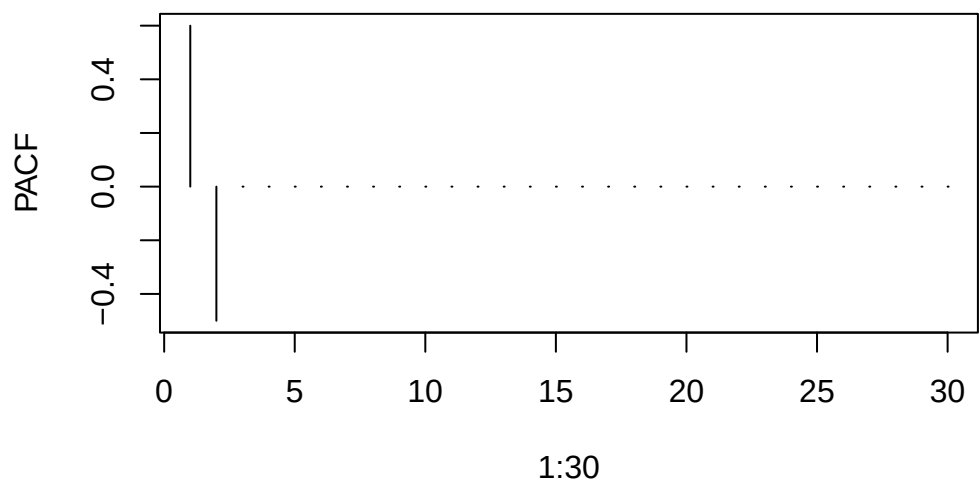


II

```
## Theoretical autocorrelations for  $X_t = 0.9$ 
##  $X_{t-1} - 0.5 X_{t-2} + e_t$ 
plot(0:30, ARMAacf(ar = c(0.9, -0.5), lag.max = 30), type = "h", ylab = "ACF")
```

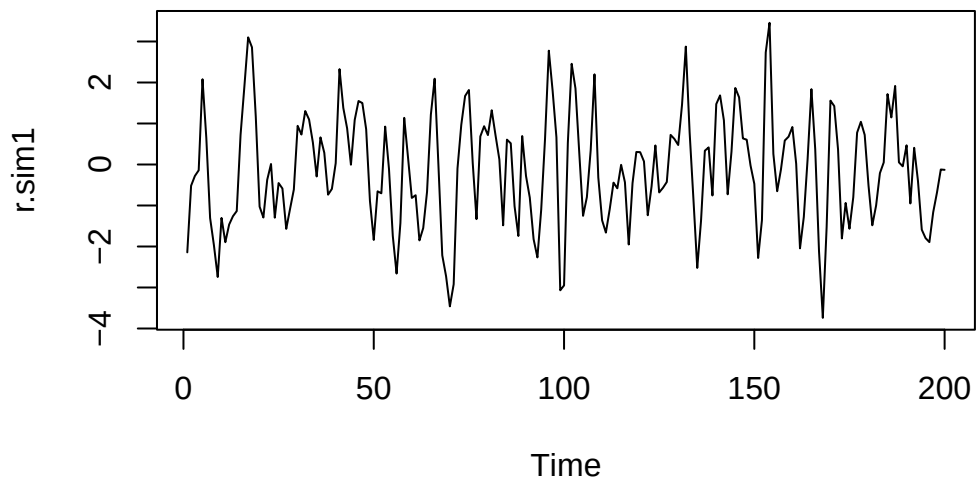


```
## Theoretical partial autocorrelations
plot(1:30, ARMAacf(ar = c(0.9, -0.5), lag.max = 30, pacf = TRUE), type = "h", ylab = "PACF")
```



III

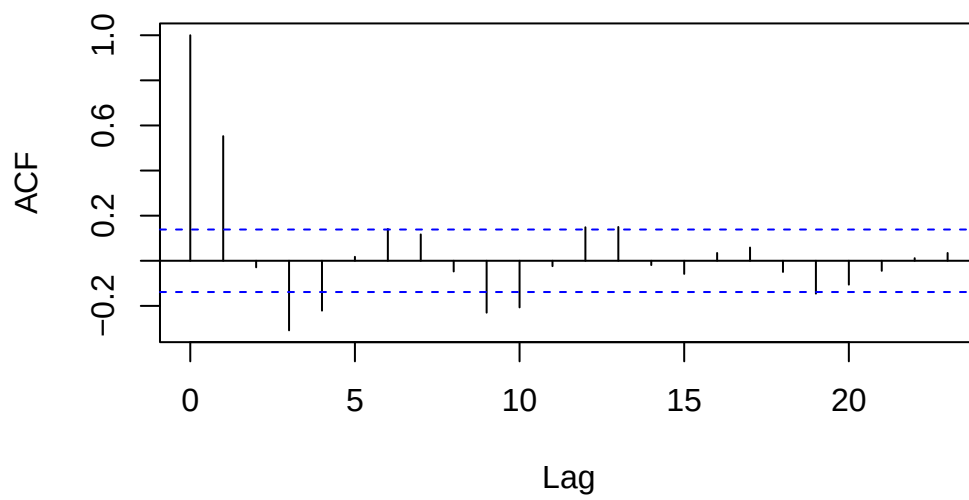
```
r.sim1 <- arima.sim(n = 200, model = list(ar = c(0.9, -0.5)))  
plot(r.sim1)
```



IV

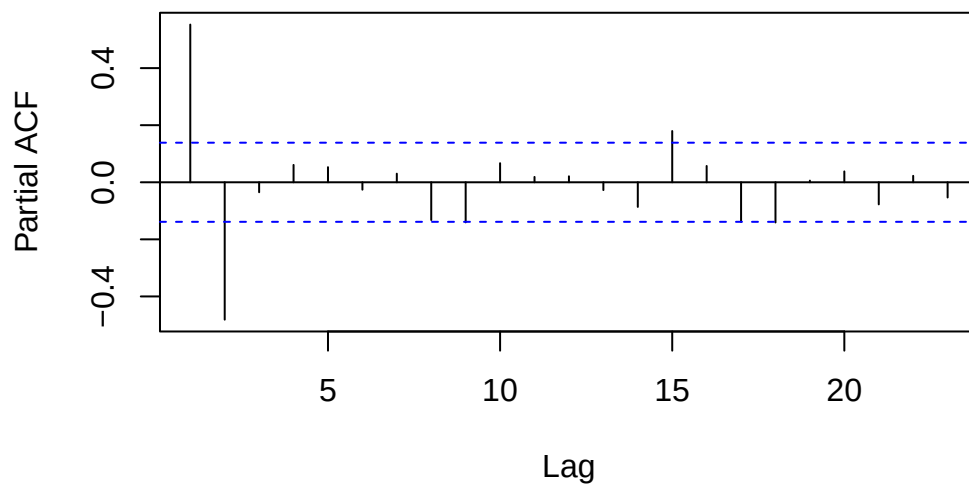
```
acf(r.sim1)
```

Series r.sim1



```
pacf(r.sim1)
```

Series r.sim1



Exercise 5.2

a

I

```
polyroot(c(1, 0.5, 2))
```

```
[1] -0.125+0.6959705i -0.125-0.6959705i
```

The process is not stationary

II

The process is not stationary since $a_1 = 1$.

b

```
polyroot(c(1, 0.5, 0.26))
```

```
[1] -0.9615385+1.709268i -0.9615385-1.709268i
```

a_2 has to be < 0.25 .

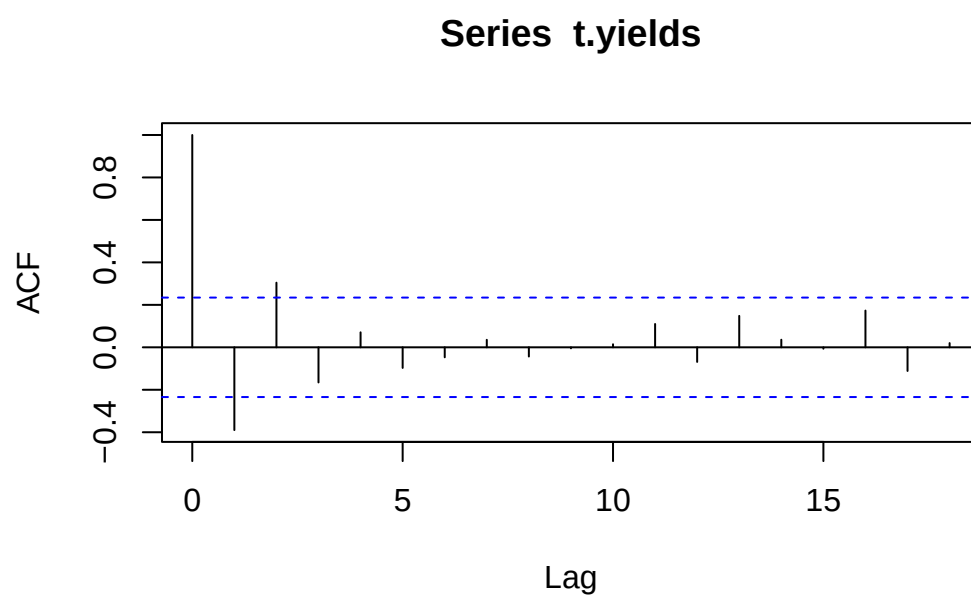
Series 6

Exercise 6.1

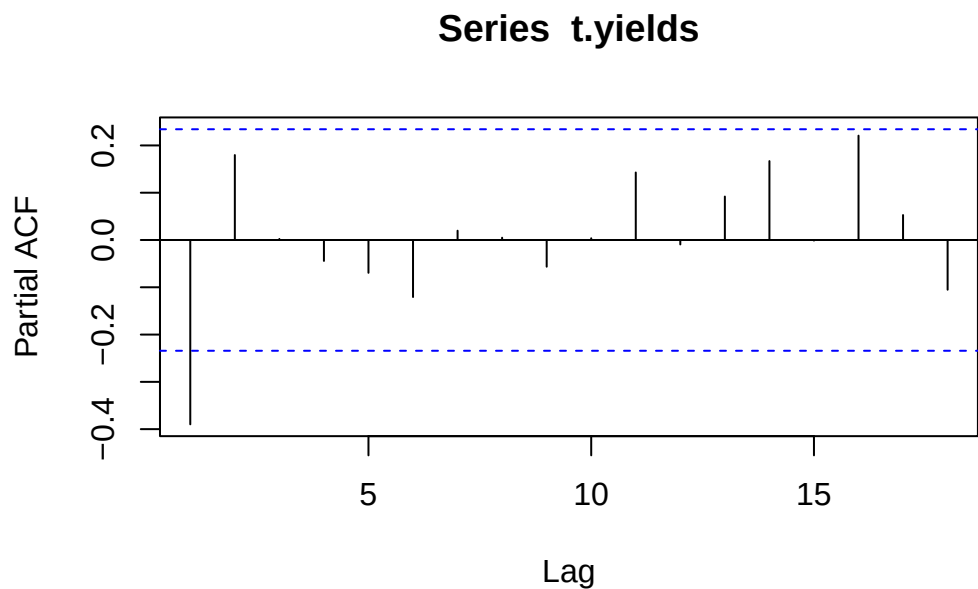
```
yields <- read.table("./data/yields.dat", header = FALSE)
t.yields <- ts(yields[, 1])
```

a

```
acf(t.yields)
```



```
pacf(t.yields)
```



The process can be modeled by a $AR(2)$.

b

```
r.yw <- ar(t.yields, method = "yw", order.max = 2)
r.yw$resid
```

Time Series:

Start = 1

End = 70

Frequency = 1

[1]	NA	NA	-23.2701676	8.5624227	-1.7185396	5.1017154
[7]	10.3471780	-11.2034906	3.9364420	1.2089867	17.4563283	-9.2111662
[13]	-2.8577303	-6.3524133	2.2572167	-2.9311234	25.3567725	14.3859910
[19]	-18.0787717	17.6571812	9.7250352	1.7201979	0.7723057	7.4553688
[25]	-3.0885436	2.3171776	1.8505361	-7.5446312	-27.8857718	0.6164494
[31]	6.0842678	18.0959587	11.4294237	20.8583891	5.3106389	2.4003827
[37]	-3.7281741	-0.3234123	-13.1089096	-2.4829289	0.5084360	2.3548536
[43]	-4.3282096	3.2157029	1.8505361	9.4553688	-3.4489132	8.6379522
[49]	-2.7310524	-4.0412731	-11.3891299	3.5161116	8.7481019	-10.3049653
[55]	-2.0635580	0.2900955	-18.1456261	-21.2240340	0.7913572	-2.3118589

```
[61] 14.3954080 -6.6314966 -8.3591694 10.8697651 -9.0885436 2.3982863
[67] -6.4316031 0.8978066 6.7490614 -28.2653704
```

```
str(r.yw)
```

List of 15

```
$ order      : int 2
$ ar         : num [1:2] -0.32 0.18
$ var.pred   : num 120
$ x.mean     : num 51.1
$ aic        : Named num [1:3] 9.84 0.298 0
..- attr(*, "names")= chr [1:3] "0" "1" "2"
$ n.used     : int 70
$ n.obs      : int 70
$ order.max   : num 2
$ partialacf : num [1:2, 1, 1] -0.39 0.18
$ resid      : Time-Series [1:70] from 1 to 70: NA NA -23.27 8.56 -1.72 ...
$ method     : chr "Yule-Walker"
$ series     : chr "t.yields"
$ frequency   : num 1
$ call       : language ar(x = t.yields, order.max = 2, method = "yw")
$ asy.var.coef: num [1:2, 1:2] 0.01444 0.00563 0.00563 0.01444
- attr(*, "class")= chr "ar"
```

c

```
r.burg <- ar(t.yields, method = "burg", order.max = 2)
r.burg$resid
```

Time Series:

Start = 1

End = 70

Frequency = 1

```
[1]      NA      NA -23.0964972  8.1701783 -1.3785617  4.8728473
[7] 10.5505765 -11.1978722  3.7977236  1.3403098 17.3914298 -8.9537665
[13] -3.1238980 -6.2261379  2.0994588 -2.8089304 25.2454443 14.7685021
[19] -18.1260033 17.4687299 10.0560689  1.6430522  0.8457311  7.4219430
[25] -2.9745155  2.2116763  1.9437802 -7.5780570 -27.9582418  0.3118307
[31]  6.2684408 18.0559266 11.6802172 20.8533189  5.5789133  2.3108038
```

```
[37] -3.6390123 -0.4223073 -13.0529651 -2.6805103 0.5941087 2.3064707
[43] -4.2697412 3.1267173 1.9437802 9.4219430 -3.3100186 8.5134115
[49] -2.5474727 -4.1848533 -11.3514455 3.3500029 8.8893344 -10.2828313
[55] -2.2022764 0.3468193 -18.1782727 -21.4299662 0.6474050 -2.2228832
[61] 14.3097254 -6.4014872 -8.5781282 10.8990986 -8.9745155 2.2181858
[67] -6.2936738 0.7334424 6.8588212 -28.2490633
```

```
str(r.burg)
```

List of 15

```
$ order      : int 2
$ ar         : num [1:2] -0.332 0.183
$ var.pred   : num 113
$ x.mean     : num 51.1
$ aic        : Named num [1:3] 11.039 0.385 0
  ..- attr(*, "names")= chr [1:3] "0" "1" "2"
$ n.used     : int 70
$ n.obs      : int 70
$ order.max  : num 2
$ partialacf : num [1:2, 1, 1] -0.407 0.183
$ resid      : Time-Series [1:70] from 1 to 70: NA NA -23.1 8.17 -1.38 ...
$ method     : chr "Burg"
$ series     : chr "t.yields"
$ frequency  : num 1
$ call       : language ar(x = t.yields, order.max = 2, method = "burg")
$ asy.var.coef: num [1:2, 1:2] 0.0136 0.0053 0.0053 0.0136
- attr(*, "class")= chr "ar"
```

d

```
r.mle <- ar(t.yields, method = "mle", order.max = 2)
r.mle$resid
```

Time Series:

Start = 1

End = 70

Frequency = 1

```
[1]      NA      NA -23.083437837  7.762346509 -1.201980697
[6]  4.561565507 10.602738000 -11.334850093  3.581349256  1.337116357
[11] 17.216926017 -8.885802464 -3.460572312 -6.220191632  1.865930909
```

```

[16] -2.816243379 25.041428463 14.930022474 -18.332496024 17.218498832
[21] 10.197216932 1.428770775 0.782162438 7.273056970 -3.008346873
[26] 2.007435328 1.906279662 -7.726943030 -28.119035966 0.003476264
[31] 6.334762287 17.898351229 11.739491795 20.694392418 5.637512389
[36] 2.088058169 -3.689772086 -0.617858931 -13.115858157 -2.934860904
[41] 0.570317679 2.153690370 -4.337204163 2.944199681 1.906279662
[46] 9.273056970 -3.326921661 8.292016980 -2.521379225 -4.423401367
[51] -11.422577628 3.121270050 8.899174969 -10.398085740 -2.418650744
[56] 0.292840720 -18.318244153 -21.679521763 0.471483903 -2.242329450
[61] 14.131552552 -6.345881671 -8.876375881 10.832761513 -9.008346873
[66] 1.963159691 -6.288177902 0.491225167 6.843044015 -28.364092605

```

```
str(r.mle)
```

List of 15

```

$ order      : int 2
$ ar         : num [1:2] -0.341 0.187
$ var.pred   : num 113
$ x.mean     : num 51.2
$ aic        : Named num [1:3] 10.808 0.301 0
  ..- attr(*, "names")= chr [1:3] "0" "1" "2"
$ n.used     : int 70
$ n.obs      : int 70
$ order.max  : num 2
$ partialacf : NULL
$ resid      : Time-Series [1:70] from 1 to 70: NA NA -23.08 7.76 -1.2 ...
$ method     : chr "MLE"
$ series     : chr "t.yields"
$ frequency  : num 1
$ call       : language ar(x = t.yields, order.max = 2, method = "mle")
$ asy.var.coef: num [1:2, 1:2] 0.0136 0.0053 0.0053 0.0136
- attr(*, "class")= chr "ar"

```

Exercise 6.2

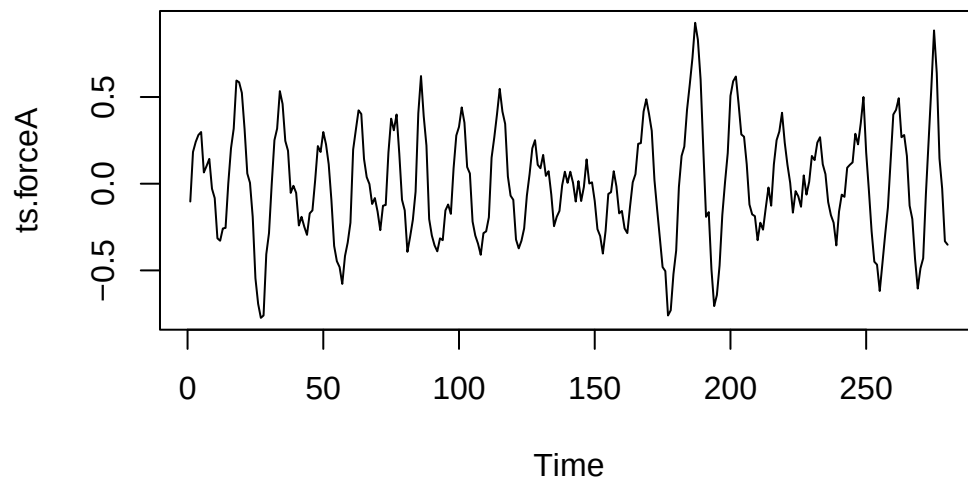
```

d.force <- read.table("./data/kraft.dat", header = FALSE)
ts.force <- ts(d.force[, 1])

```

a

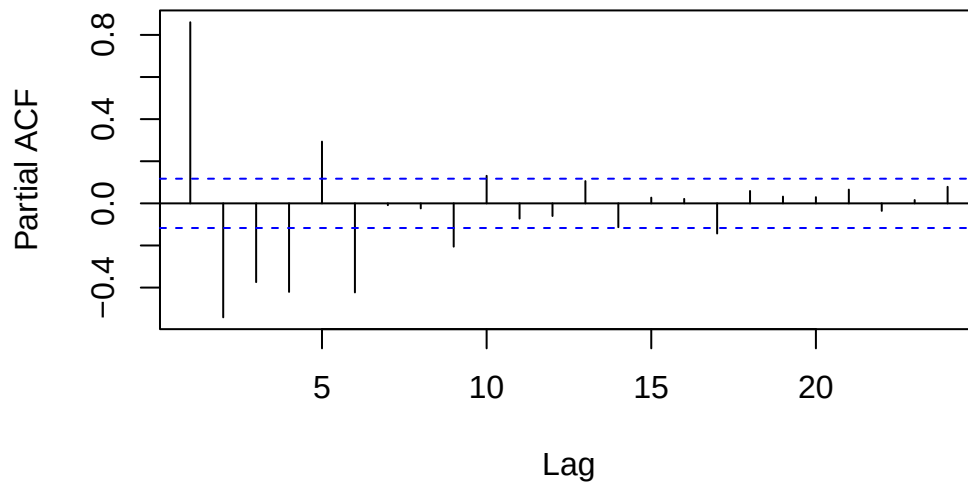
```
ts.forceA <- window(ts.force, end = 280)  
plot(ts.forceA)
```



b

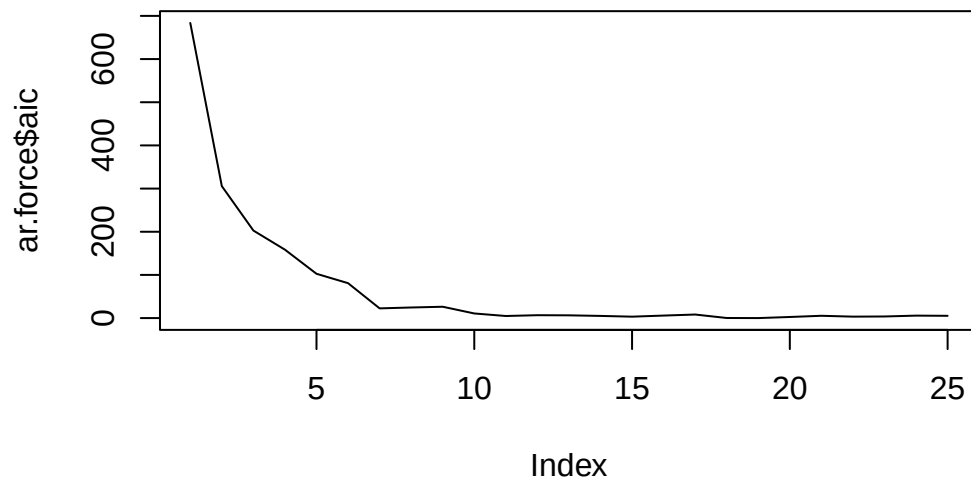
```
pacf(ts.forceA)
```

Series ts.forceA



We should use a $AR(7)$ model.

```
ar.force <- ar(ts.forceA, method = "ols")  
plot(ar.force$aic, type="l")
```



c

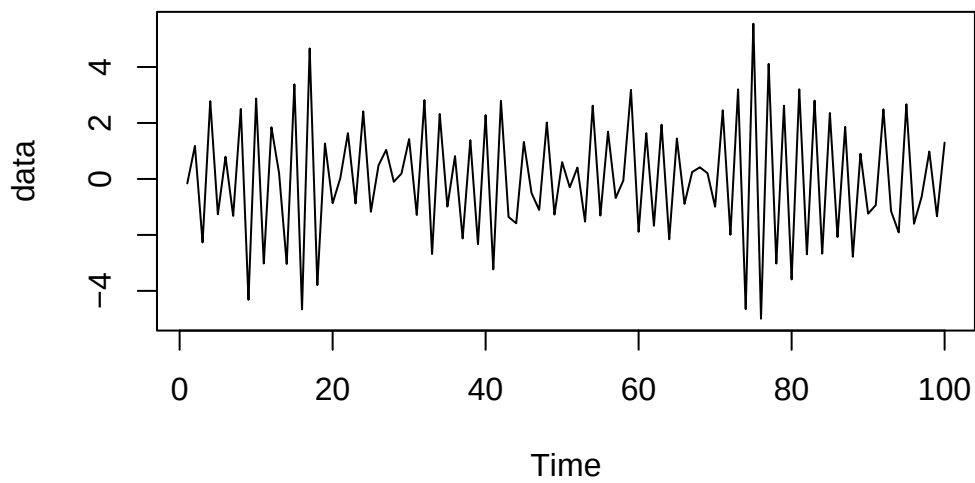
```
ar.force <- arima(ts.forceA, order = c(7, 0, 0), method = "ML")
```

Series 7

Exercise 7.1

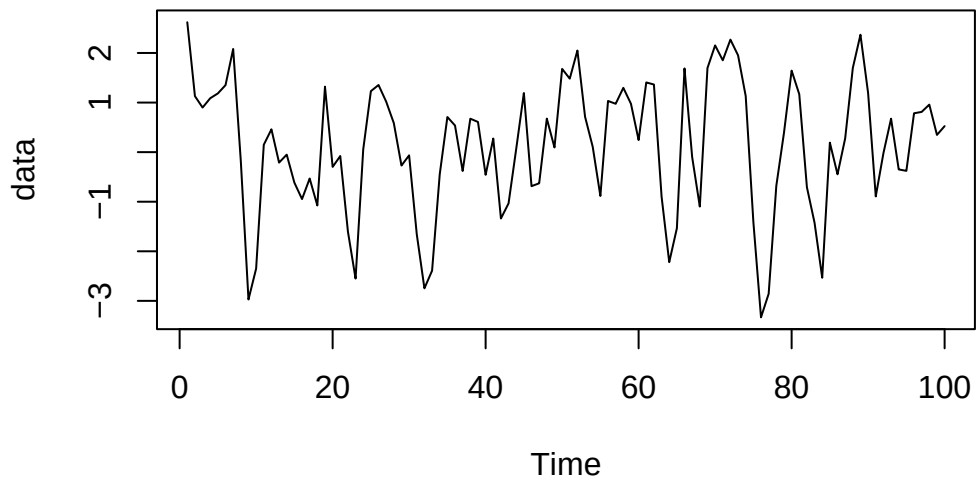
a

```
data <- arima.sim(n = 100, model = list(ar = c(-0.75), ma = c(-0.3, 0.25)))  
plot(data, type="l")
```



b

```
data <- arima.sim(n = 100, model = list(ar = c(0.5, -0.3), ma = c(0.25)))  
plot(data, type="l")
```

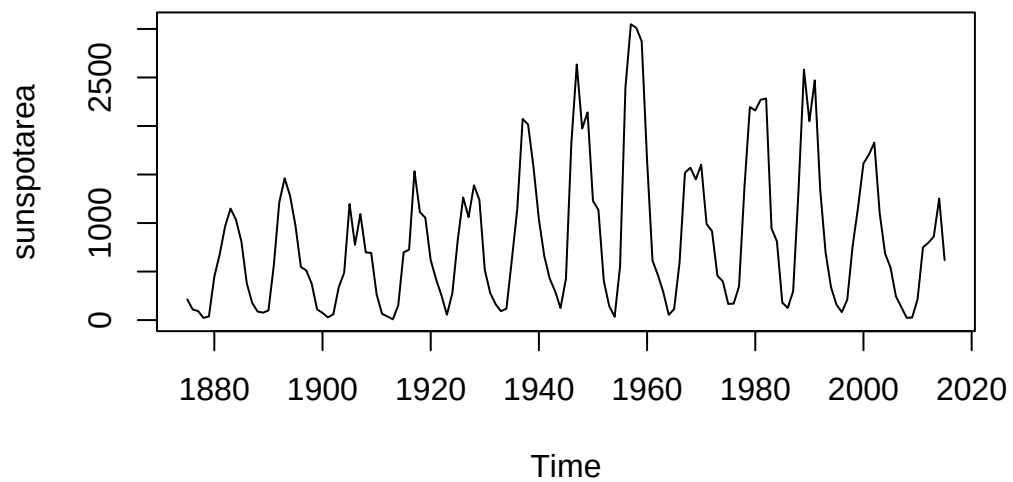


Exercise 7.2

a

```
library(fpp2)
data(sunspotarea)

plot(sunspotarea, type="l")
```



We should log-transform because the data are not stationary.

b

```
subset <- log(sunspotarea[1:100])
ar.sun<- ar(subset, method = "ols")
ar.sun
```

Call:

```
ar(x = subset, method = "ols")
```

Coefficients:

1	2	3	4	5	6	7	8
0.6986	-0.1578	-0.0555	-0.0398	0.0387	-0.1249	0.1495	-0.2127
9	10						
0.1133	0.3531						

Intercept: 0.0581 (0.06088)

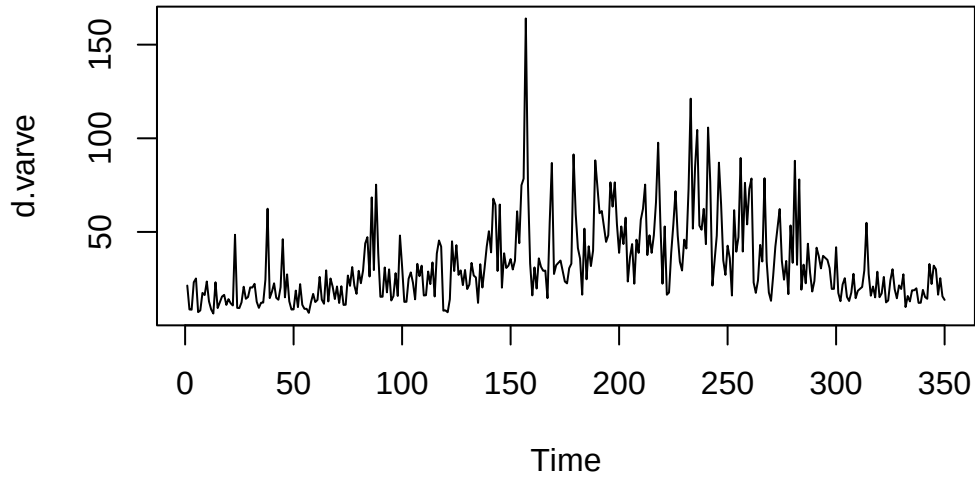
Order selected 10 sigma^2 estimated as 0.3287

Exercise 7.3

```
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/varve.dat"  
d.varve <- ts(scan(t.url)[201:550])
```

a

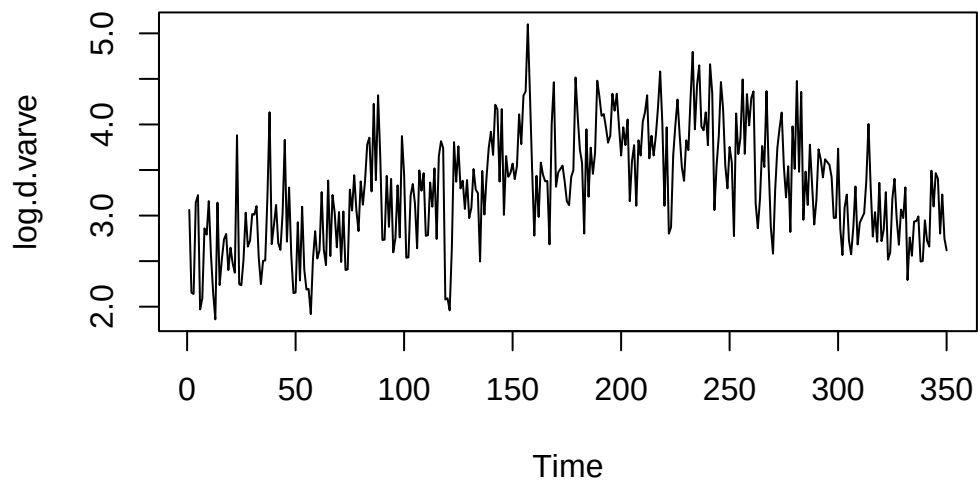
```
plot(d.varve, type="l")
```



We should log-transform the data because the process is not stationary.

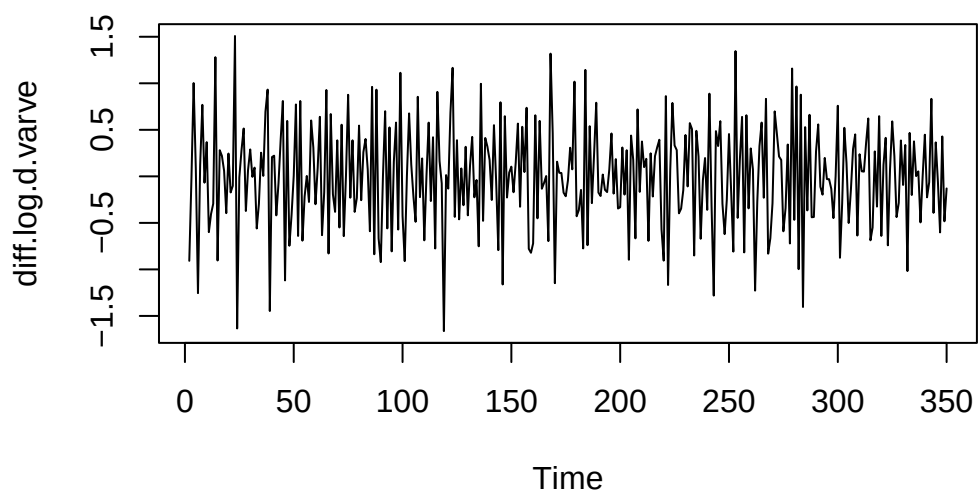
b

```
log.d.varve <- log(d.varve)  
plot(log.d.varve, type="l")
```



Still not stationary.

```
diff.log.d.varve <- diff(log.d.varve)
plot(diff.log.d.varve, type="l")
```



By applying differencing, we can make the process stationary.

Series 8

Exercise 8.1

a

```
arima.sim(model = list(order=c(2, 1, 2), ar=c(0.5, 0.5), ma=c(-0.4, 0.3)), n=100)
```

```
Error in arima.sim(model = list(order = c(2, 1, 2), ar = c(0.5, 0.5),: 'ar'
part of model is not stationary
```

b

1. Substitute the given coefficients:

$$(1 - 0.5B - 0.5B^2)(1 - B)X_t = (1 - 0.4B + 0.3B^2)E_t$$

2. Expand the left-hand side (AR polynomial multiplied by the difference operator):

$$(1 - B - 0.5B + 0.5B^2 - 0.5B^2 + 0.5B^3)X_t = (1 - 0.4B + 0.3B^2)E_t$$

3. Simplify the expression:

$$(1 - 1.5B + 0.5B^3)X_t = (1 - 0.4B + 0.3B^2)E_t$$

The resulting expression represents an **ARMA**(3, 2) process with the following parameters:

- Autoregressive (AR) Coefficients:

- $\alpha_1^* = 1.5$
- $\alpha_2^* = 0$
- $\alpha_3^* = -0.5$

- Moving Average (MA) Coefficients:

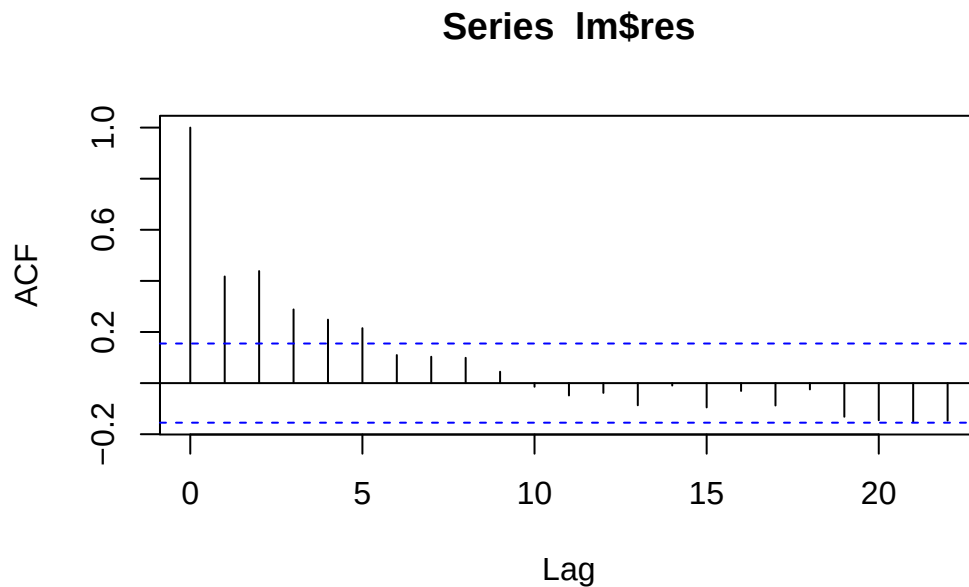
- $\beta_1 = -0.4$
- $\beta_2 = 0.3$

Exercise 8.2

```
d.beluga <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/beluga.dat", header = TRUE)
d.beluga <- ts(d.beluga)
```

a

```
lm <- lm(NURSING ~ PERIOD + BOUTS + LOCKONS + DAYNIGHT, data=d.beluga)
acf(lm$res)
```



looks more or less good.

b

```
r.burg <- ar(d.beluga, method = "burg")
str(r.burg)
```

List of 14

```
$ order      : int 6
$ ar          : num [1:6, 1:5, 1:5] 2.5182 -1.6528 -0.0945 -0.095 0.4912 ...
..- attr(*, "dimnames")=List of 3
.. ..$ : chr [1:6] "1" "2" "3" "4" ...
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
$ var.pred   : num [1:5, 1:5] 6.17e-04 -7.08e-04 -3.09e-03 -1.23e-03 8.31e-05 ...
..- attr(*, "dimnames")=List of 2
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
$ x.mean     : Named num [1:5] 80.5 2.79 11.49 3.61 0.5
..- attr(*, "names")= chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
$ aic        : Named num [1:23] 3946 2628 242 157 130 ...
..- attr(*, "names")= chr [1:23] "0" "1" "2" "3" ...
$ n.used     : int 160
$ n.obs      : int 160
$ order.max  : num 22
$ partialacf: num [1:22, 1:5, 1:5] 0.9997 -0.9993 0.762 0.3808 0.0846 ...
..- attr(*, "dimnames")=List of 3
.. ..$ : chr [1:22] "1" "2" "3" "4" ...
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
$ resid      : 'mts' num [1:160, 1:5] NA NA NA NA NA ...
..- attr(*, "tsp")= num [1:3] 1 160 1
..- attr(*, "dimnames")=List of 2
.. ..$ : NULL
.. ..$ : chr [1:5] "PERIOD" "BOUTS" "NURSING" "LOCKONS" ...
$ method     : chr "Burg"
$ series     : chr "d.beluga"
$ frequency  : num 1
$ call       : language ar(x = d.beluga, method = "burg")
- attr(*, "class")= chr "ar"
```

c

```
library(nlme) # Load the package containing the procedure gls()
r.bel.gls <- gls(NURSING ~ BOUTS + LOCKONS + DAYNIGHT + PERIOD, data = d.beluga, correlation
summary(r.bel.gls)
```

Generalized least squares fit by maximum likelihood
 Model: NURSING ~ BOUTS + LOCKONS + DAYNIGHT + PERIOD
 Data: d.beluga
 AIC BIC logLik
 565.0884 601.9905 -270.5442

Correlation Structure: ARMA(6,0)
 Formula: ~PERIOD
 Parameter estimate(s):

	Phi1	Phi2	Phi3	Phi4	Phi5	Phi6
	0.26391489	0.33076391	-0.01473528	0.08533177	0.11805117	-0.08347252

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	1.5947050	0.8171076	1.951646	0.0528
BOUTS	0.1634638	0.3265945	0.500510	0.6174
LOCKONS	2.5699567	0.1888387	13.609269	0.0000
DAYNIGHT	-0.2836475	0.1727329	-1.642116	0.1026
PERIOD	0.0032607	0.0071800	0.454133	0.6504

Correlation:

	(Intr)	BOUTS	LOCKON	DAYNIG
BOUTS	-0.271			
LOCKONS	-0.082	-0.810		
DAYNIGHT	-0.038	-0.119	0.060	
PERIOD	-0.652	-0.207	0.206	0.020

Standardized residuals:

	Min	Q1	Med	Q3	Max
	-2.69029710	-0.55948989	0.02751544	0.62878372	2.48793230

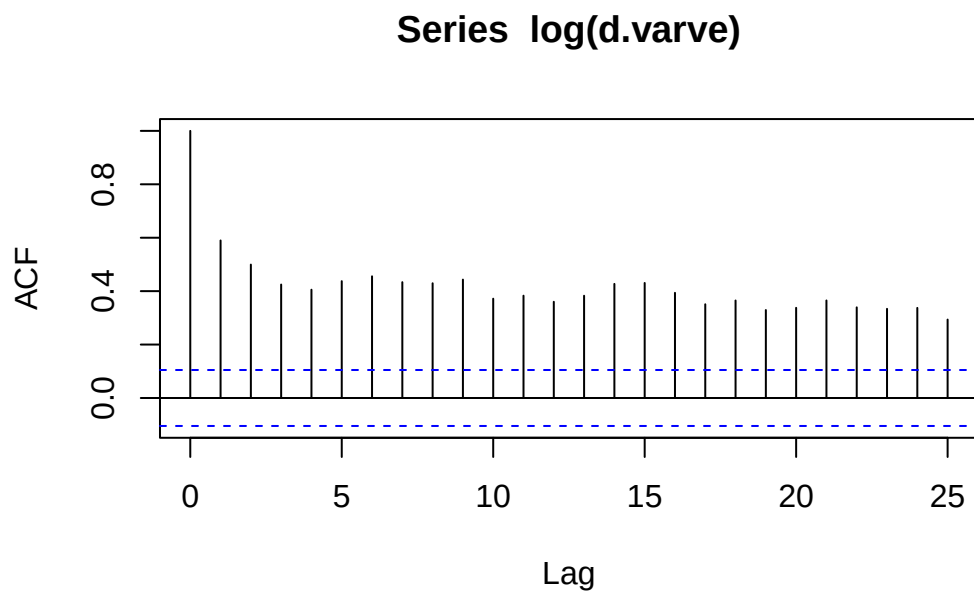
Residual standard error: 1.595643
 Degrees of freedom: 160 total; 155 residual

```
d.resid <- ts(resid(r.bel.gls))
```

Exercise 8.3

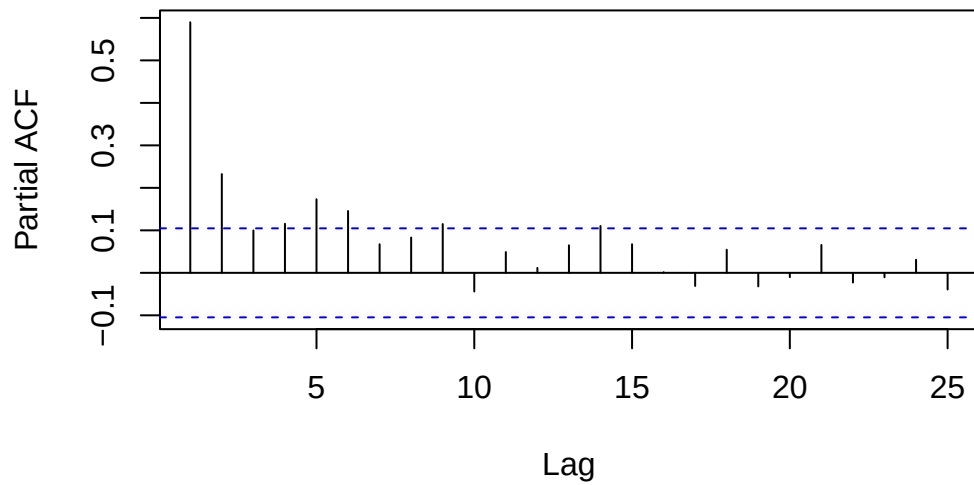
a

```
acf(log(d.varve))
```



```
pacf(log(d.varve))
```

Series log(d.varve)



```
r.varve <- arima(log(d.varve), order = c(0, 0, 9))
```

b

```
r.varve
```

Call:

```
arima(x = log(d.varve), order = c(0, 0, 9))
```

Coefficients:

	ma1	ma2	ma3	ma4	ma5	ma6	ma7	ma8	ma9
	0.4074	0.3063	0.2006	0.1610	0.1934	0.2236	0.1694	0.1453	0.1674
s.e.	0.0536	0.0572	0.0594	0.0612	0.0587	0.0593	0.0561	0.0530	0.0522
intercept									
	3.2892								
s.e.	0.0753								

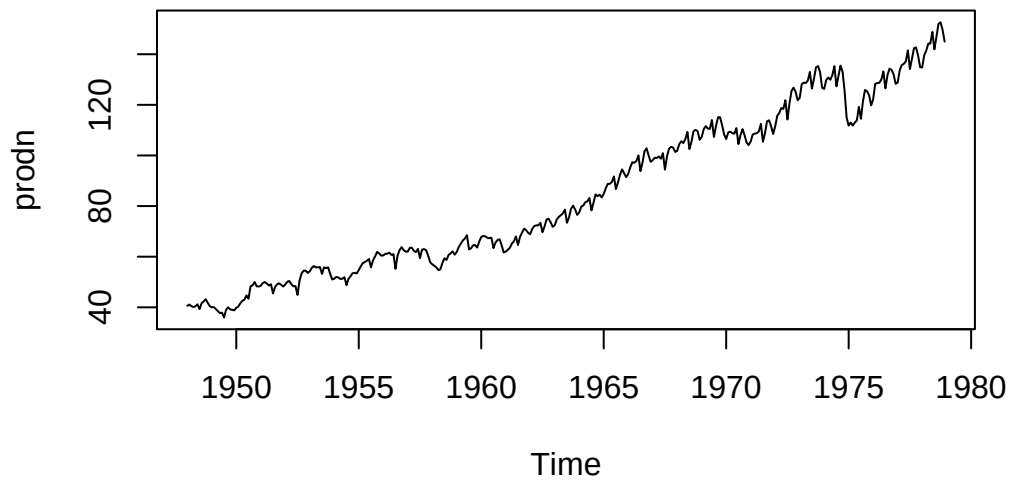
sigma^2 estimated as 0.228: log likelihood = -238.25, aic = 498.49

Series 9

Exercise 9.1

a

```
library(astsa)
data(prodn)
plot(prodn, type="l")
```

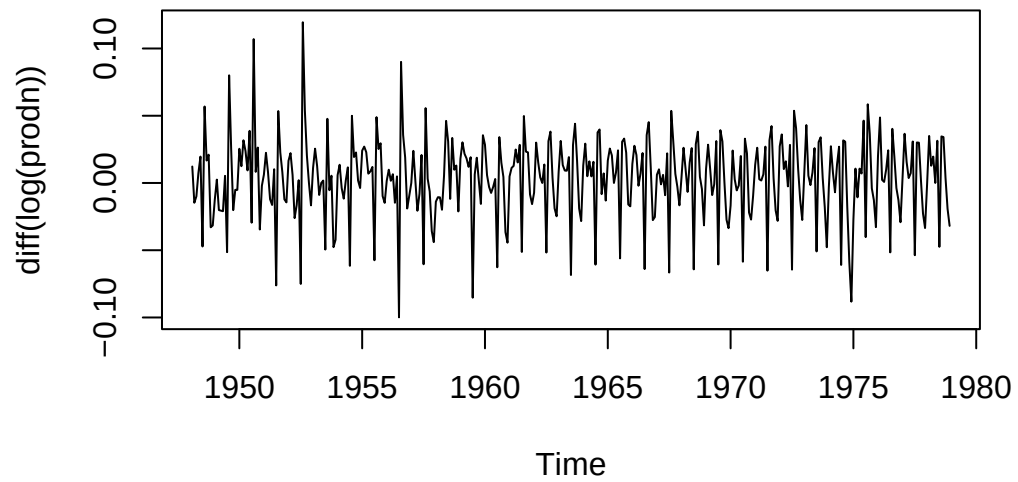


No constance variance. The process is not stationary.

b

We can apply tome transformations

```
plot(diff(log(prodn)))
```



c

```
arima(prodn, order = c(1, 1, 1), seasonal = c(1, 1, 1))
```

Call:

```
arima(x = prodn, order = c(1, 1, 1), seasonal = c(1, 1, 1))
```

Coefficients:

	ar1	ma1	sar1	sma1
	0.5767	-0.2783	-0.0352	-0.6700
s.e.	0.1178	0.1376	0.0664	0.0424

sigma^2 estimated as 1.426: log likelihood = -577.01, aic = 1164.02

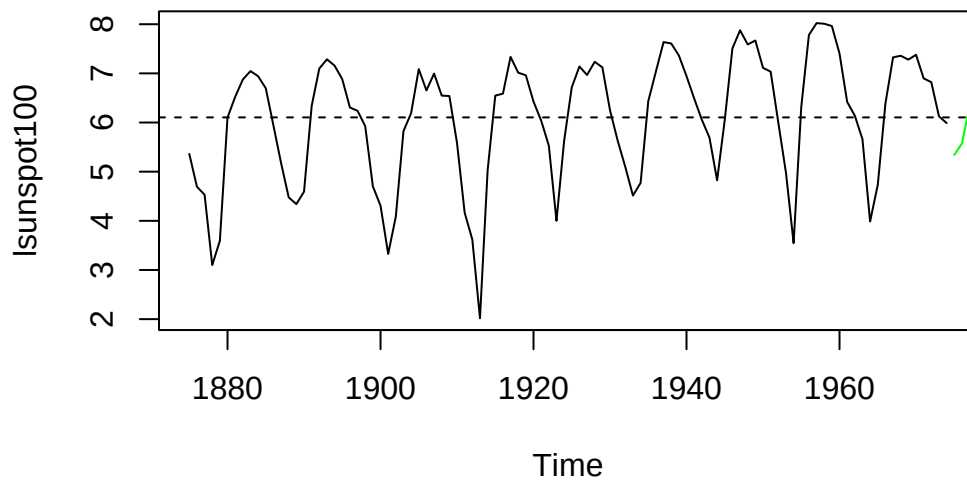
Series 10

Exercise 10.1

```
library(fpp2)
lsunspot100 <- window(log(sunspotarea), start = 1875, end = 1974)
fit.ar10 <- arima(lsunspot100, order = c(10, 0, 0))
```

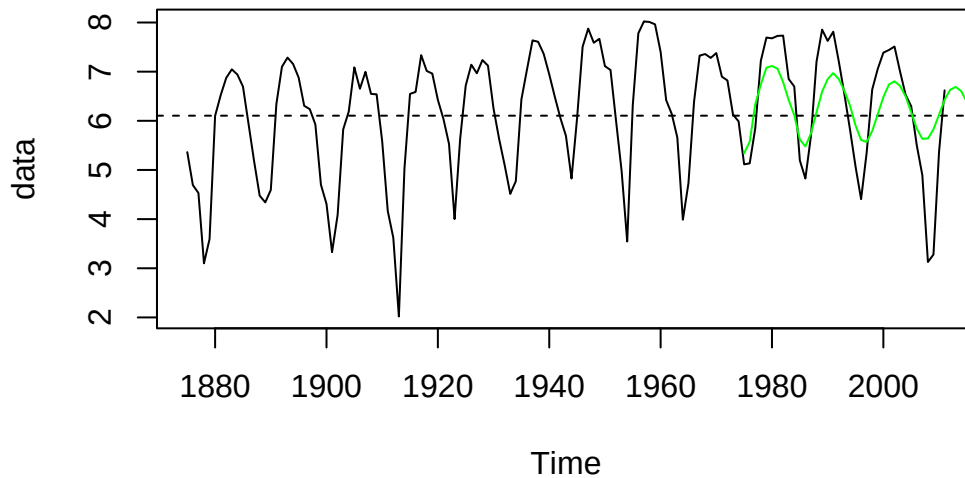
a

```
pred <- predict(fit.ar10, n.ahead = 100)
plot(lsunspot100)
abline(h = fit.ar10$coef["intercept"], lty = 2)
lines(pred$pred, col = "green")
```



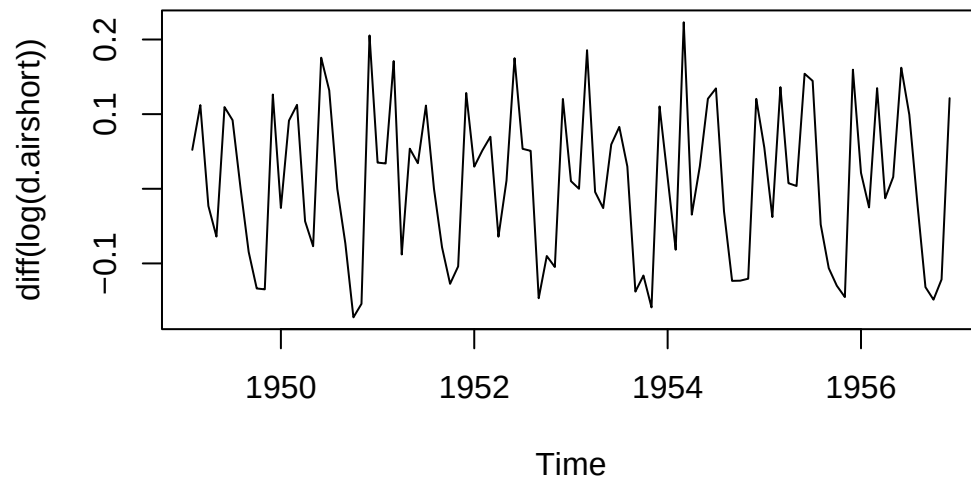
b

```
data <- lsunspot100 <- window(log(sunspotarea), start = 1875, end = 2011)
plot(data)
abline(h = fit.ar10$coef["intercept"], lty = 2)
lines(pred$pred, col = "green")
```



Exercise 10.2

```
d.air <- AirPassengers
d.airshort <- window(d.air, end = c(1956, 12))
plot(diff(log(d.airshort)))
```



a

```
log.airshort <- log(d.airshort)
fit.sarima <- arima(log.airshort,
                    order = c(0, 1, 1),
                    seasonal = list(
                      order = c(0, 1, 1),
                      period = 12
                    )
)
pred.sarima <- predict(fit.sarima, n.ahead = 48)

plot(log.airshort)
abline(h = fit.sarima$coef["intercept"], lty = 2)
lines(pred.sarima$pred, col = "green")
```

